

Anti-atherosclerotic effects of garlic preparation in freeze injury model of atherosclerosis in cholesterol-fed rabbits



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ABSTRACT

Background: Garlic (*Allium sativum* L.) is one of the most popular substances used to reduce various risks associated with cardiovascular disease. However, little is known on the direct effects of garlic on atherosclerosis.

Purpose: In the present study we have examined the effect of per oral administration of the time-released garlic herbal preparation on serum atherogenicity and formation of intimal thickening after freeze injury in cholesterol-fed rabbits.

Methods: Group 1 rabbits maintained on the standard cholesterol-rich diet served as the control. Group 2 rabbits were fed the cholesterol-rich diet and treated with garlic preparation containing 300 mg garlic powder.

Results: Local thickening of the aortic media (i.e., the neointima formation) in the freeze injury zone was observed in all the rabbits. Regular garlic preparation therapy prevented the neointima formation and the accumulation of free and esterified cholesterol, triglycerides, phospholipids and collagen in the neointima, the effects being statistically significant. Garlic preparation also decreased serum lipid content by 1.5-fold and lowered atherogenic activity of blood serum (ability to induce lipid accumulation in cultured cells) induced by cholesterol-rich diet.

Conclusion: The results obtained indicate that garlic preparation prevents the development of cholesterol-induced experimental atherosclerosis and possesses the direct anti-atherogenic activity.

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Introduction

Garlic (*Allium sativum* L.) is one of the most popular substances used to reduce various risks associated with cardiovascular disease. However, little is known on the direct effects of garlic on atherosclerosis.

Atherosclerosis-related effects of garlic and garlic-derived preparations may be realized at the arterial level, preventing new and suppressing the existing atherosclerotic. The mechanisms of these effects have been shown to be associated with the effects of the garlic components on blood lipid profile. Recently, using an animal experimental model, Mohammadi and Oshaghi (2014) studied effects

of garlic on lipid profile as well as LXR α expression in intestine and liver. The study found that garlic extract reduced LXR α expression in the liver and increased its expression in the intestine. The study of Mohammadi and Oshaghi (2014) demonstrated a role of garlic in reducing serum triglyceride and cholesterol. It seems that garlic could affect directly intimal vascular cells. Supporting this possibility is a study of Orekhov et al. (1996). Which demonstrated that water extract of garlic powder produces the direct anti-atherogenic effect on cultured smooth muscle cells isolated from grossly normal human aortic intima by preventing the intracellular accumulation of cholesterol and reducing cellular proliferation stimulated by atherogenic serum. Many studies reported anti-atherogenic effects of garlic preparations observed *in vivo* (Bordia et al., 1975; Jain, 1975; Kritchevsky, 1975; Jain and Konar, 1978; Mirhardi et al., 1991; Rafieian-Kopaei et al., 2013).

In the present study we have examined the effect of orally administered garlic powder time-released garlic preparation containing

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300 mg dried garlic powder from *Allium sativum* L. bulbs (Allicor, INAT-Pharma, Russia) on the atherogenic properties of blood serum and on the formation of intimal thickening in a freeze-injured aorta of cholesterol-fed rabbits. Long-acting garlic preparations have not been examined previously for their anti-atherogenic effect in the animal model.

Material and methods

Aorta injury model

The experiment procedures carried out in the present study complied with the “Animal Research: Reporting *In Vivo* Experiments” (ARRIVE) guidelines. The animal studies were carried out in accordance with the Guide for the Care and Use of Laboratory Animals, Institute of Experimental Cardiology, Russian Cardiology Research Center; Moscow, Russia. The study was approved by the Ethical committees of Institute of Atherosclerosis Research, Moscow, and by Russian Cardiology Research Center, Moscow.

Commercially available garlic preparation was used in this study (Allicor, INAT-Pharma, Russia; batch no. 13213, quality control protocol no. 172/13). The single herbal constituent of Allicor was dried garlic powder from bulbs of *Allium sativum* L. The plant name has been checked with <http://www.theplantlist.org> (*Allium sativum* L.; Status: accepted; Confidence level: ***; Source: WCSP; Date supplied: 2012-03-23). One tablet contained 300 mg dried garlic powder standardized by allicin content, the most recommended bioavailable marker compound to ensure a consistent quality and reproducible pharmacological activity (Borlinghaus et al., 2014). In brief, 150 mg powdered tablets were extracted with 2.5 ml water for 20 min., and further with 150 μ l 40% NaOH for 3 min., and neutralized by adding 475 ± 25 μ l 10% HCl and 500 μ l phosphate buffer (pH 7.0). Then methanol was added to final volume of 10 ml, and the mixture was centrifuged to obtain clear supernatant. The content of active ingredient in the final product was determined by measuring free allicin by HPLC. Allicin content accounted for 0.7%, or 2.1 mg per tablet. The fingerprint of extract sample obtained by HPLC analysis is shown at Fig. 1. According to manufacturer's specification, the delayed disintegration of tablets and, thus, the prolonged biological effect of garlic preparation was provided by polymeric matrix containing dried garlic particles.

Experiments were performed on male Chinchilla rabbits, aged 12–15 weeks weighing 3.0–3.5 kg. The animals were maintained on the standard diet. Rabbits were anesthetized with Nembutal (40 mg/kg). Rabbit aorta was denuded by freeze injury (Malczak and Buck, 1977). A 12 mm aluminium bar, cooled down to the temperature of liquid nitrogen was placed on the exterior of the abdominal aorta (infrarenal segment) for 1 min.

Twelve weeks after the procedure, the rabbits were divided into two groups of 12 animals each. Group 1 rabbits (control) were given cholesterol (Fluca Chemie, AG, Buchs, Switzerland) perorally in dose

of 2 g. Group 2 rabbits received cholesterol in the same dose and commercially available garlic preparation containing 300 mg dried garlic powder standardized by allicin content (0.7%) (Allicor, INAT-Pharma, Russia). Cholesterol and garlic preparation were mixed with vegetable oil and given to the rabbits once daily for 35 days.

Blood was drawn before the aorta denudation procedure, 12 weeks after it, and on days 7 and 35 of the long-term treatment.

Light microscopy

Aortas were dissected in 17 weeks after freeze injury. The animals were sacrificed with Nembutal (80 mg/kg). For morphological studies, an aortic segment (5 mm wide) from the central part of the injury was fixed with 2.5% glutaraldehyde. The thickness of the media and intima was measured on semithin sections with an ocular micrometer (Opton-3 light microscope, Karl Zeiss, Jena, Germany). The remainder of the aorta was used in biochemical studies. Cell content in the neointima was determined by the ethanol-alkaline dissociation method as described previously (Orehov et al., 1984).

Biochemical studies

The intima and the media were separated mechanically. Lipids were extracted with a chloroform:methanol mixture (2:1) and separated by thin-layer chromatography using the following systems: (1) benzene-diethyl ether-ethanol-acetic acid (50:40:2:0.2) and, (2) petroleum ether-diethyl ether-acetic acid (90:10:1). Phospholipids, total and esterified cholesterol and triglycerides were measured densitometrically (Orehov et al., 1995). Collagen content was determined as described elsewhere (Orehov et al., 1984).

Cell culture

Rabbit blood serum was added to cultured peritoneal macrophages of BALB/c mice obtained as described (Tertov et al., 1989). Cells were washed with medium 199 and incubated for 3 h in medium 199 containing glutamine, antibiotics (all reagents – “GIBCO Europe”, Paisley, UK) and 10% of the serum tested. Cells were extensively washed and the total cholesterol content was determined as described previously (Orehov et al., 1989).

Serum lipids measurements

The total cholesterol content in blood serum samples was determined colorimetrically using a Cholesterol Monotest kit (Boehringer Mannheim, GmbH, Mannheim, Germany). The triglyceride level was evaluated colorimetrically using a Peridochrome Triglyceride kit (Boehringer Mannheim, GmbH).

Statistics

The significance of the differences was evaluated by the dispersion analysis using a BMDP statistical software (Dixon and Brown, 1977).

Results

Effect of garlic preparation on serum lipid levels

Cholesterol-rich diet caused hyperlipidemia in all rabbits. The total serum cholesterol content significantly increased from 141 ± 10 to 1533 ± 94 mg/dl ($p < 0.05$) and the triglyceride content from 129 ± 9 to 270 ± 67 mg/dl ($p < 0.05$). Garlic preparation treatment prevented the rise of serum blood lipids, which was seen after 1 and 5 weeks on cholesterol-rich diet (Fig. 2). By the end of the study (5 weeks on cholesterol-rich diet) in garlic preparation-treated animals the cholesterol and triglyceride contents were 1.2- and 1.7-fold lower,

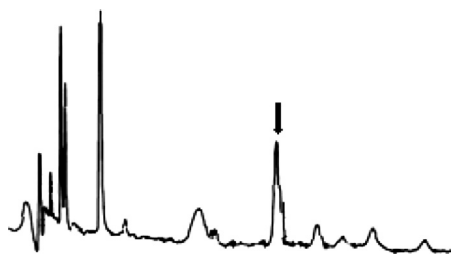


Fig. 1. The fingerprint of water extract of garlic preparation obtained by HPLC analysis. Precolumn Spherisorb ODS 2.10 μ m $\times$ 10 mm, column Spherisorb ODS 2.5 μ m $\times$ 120 mm, isocratic carrier phase methanol/water, 70:30 vol:vol, 0.5 ml/min., ultraviolet detection at 230 nm. Allicin peak is shown by arrow, retention time 10.30 min.

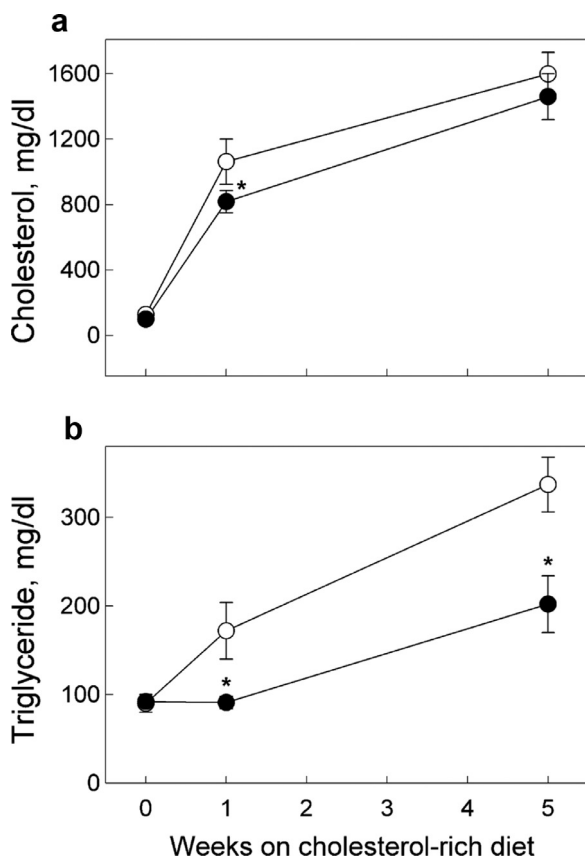


Fig. 2. The effect of garlic preparation on the lipid content in rabbit serum. a - total cholesterol; b - triglycerides. White-colored circles - serum lipid content in cholesterol-fed rabbits; black-colored circles - serum lipid content in garlic preparation-treated rabbits fed a cholesterol-rich diet. (*) - statistically significant in comparison with the control ($p < 0.05$).

respectively, than in the control group (Fig. 2a, b). However, the difference in serum cholesterol between garlic preparation-treated and control animals was significant after 1-week cholesterol-rich diet, but did not reach statistical significance after 5-week treatment.

Effect of garlic preparation on the atherogenic effects of cholesterol-rich diet

Ex vivo. Blood serum of cholesterol-fed rabbits markedly increased the total intracellular cholesterol content after 3 h incubation with cultured mouse peritoneal macrophages, i.e., exhibited an atherogenic activity (Fig. 3). Both in control and garlic preparation-treated animals, serum atherogenicity remained high throughout the entire observation period. However, garlic preparation significantly decreased the atherogenic potential of blood serum, which was induced by a cholesterol-rich diet. After 7 days of garlic preparation therapy, serum atherogenicity decreased 1.2-fold and after 35 days it was 1.5-fold lower compared with the control (Fig. 3).

In vivo. In intact rabbit aorta, the intima was not microscopically pronounced. Freeze injury caused the substantial thickening of the intima, i.e., the formation of the neointima. Local intimal thickening in the injury zone was observed 17 weeks after the procedure in all the rabbit. In garlic preparation-treated rabbits, the neointima was 1.5-fold thinner than in control rabbits (Table 1). Cell content of the intima of garlic preparation-treated rabbits was 2-fold lower than in the control (Table 1). In addition, garlic preparation significantly decreased the concentrations of free and esterified cholesterol, triglycerides, phospholipids and collagen in the neointima (Table 1).

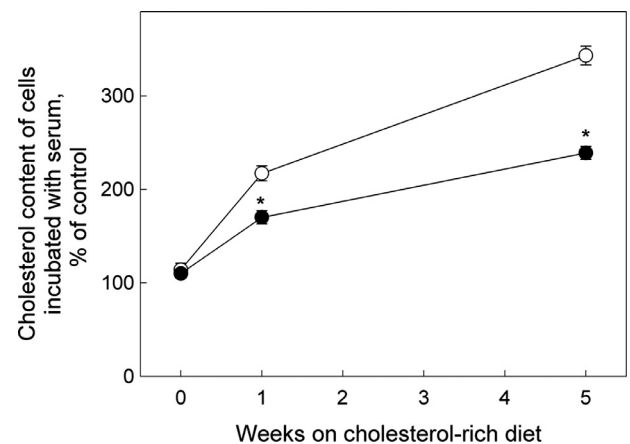


Fig. 3. The effect of garlic preparation on serum atherogenicity induced by cholesterol-rich diet identified in *in vitro* experiments. Serum was added to cultured peritoneal macrophages to a final concentration of 10%. The intracellular cholesterol content was determined 4 h afterwards. The cholesterol content in cells incubated with serum obtained before the drug administration was assumed as 100%. White-colored circles - serum atherogenicity (cholesterol-fed rabbits); black-colored circles - serum atherogenicity in garlic preparation-treated rabbits fed a cholesterol-rich diet. (*) - statistically significant in comparison with the control ($p < 0.05$).

Table 1

The effects of garlic preparation on atherogenic indices of the rabbit aorta.

Parameter	Control	Garlic preparation	% of control
Intima thickness, μm	180 ± 30	113 ± 9	63*
Intima/media index	$0.84 \pm .07$	0.64 ± 0.05	76*
Number of cells per cm^2	$(7.9 \pm 0.1) \times 10^6$	$(3.6 \pm 0.1) \times 10^6$	46*
Esterified cholesterol, mg/cm^2	5.9 ± 1.1	1.7 ± 0.4	29*
Free cholesterol, mg/cm^2	1.9 ± 0.4	0.6 ± 0.1	32*
Triglycerides, mg/cm^2	9.9 ± 2.3	2.4 ± 0.7	24*
Phospholipids, mg/cm^2	0.9 ± 0.1	0.6 ± 0.1	67*
Collagen, mg/cm^2	10.5 ± 1.5	6.1 ± 0.6	58*

* Statistically significant difference as compared to control ($p < 0.05$).

Discussion

In this study we have demonstrated the suppressing effect of long-acting garlic preparation on experimental atherosclerosis in a model of mechanical aortic injury. In the present study we induced local thickening of the rabbit aorta intima by freeze injury, since atherosclerosis is developed predominantly in the arteries with intima thickening. Intimal thickening with subsequent formation of the neointima in laboratory animals can be induced either by a balloon catheterization or freeze injury. In contrast to balloon deendothelialization, the endothelium regenerates completely after freeze injury. Additionally, the size of the freeze injury exactly corresponds to that of the applicator (Malczak and Buck, 1977; Kolpakov et al., 1993). Thus, the aortic intima thickening caused by freeze injury can be employed as a model for the investigation of the effects of various drugs on the vascular wall (Malczak and Buck, 1977; Kolpakov et al., 1993).

It should be noted that a lot of garlic-based products and preparations are present on the market now. They can be generally classified into four groups, i.e., garlic essential oil, garlic oil macerate, garlic powder and garlic extract. As compared to other garlic preparations, dehydrated garlic powder is thought to retain the same ingredients as raw garlic, both water-soluble and organic-soluble, although the proportions and amounts of various constituents may differ significantly (Iberl et al., 1990; Amagase et al., 2001). The comparative study of Allicor and Kwai in hypertensive patients (Sobenin et al., 2009) showed the differences between preparations in pharmacodynamics, although they both contain just garlic powder. The difference

was referred to time-released design of Allicor, as its biological effect lasts for 12–16 h after single dose administration (Orekhov et al., 1995; Sobenin et al., 2013; Orekhov et al., 2014). This design allows longer circulation of bioactive compounds and their remainders, as well as slower disintegration of Allicor during digestion. This process results in absorption of bioactive compounds in a dispensed way, consequently low and steady concentration of active metabolites in circulation that may allow better bioavailability of active constituents of garlic.

We have shown that garlic preparation inhibits the rise in serum cholesterol and triglycerides induced by the used cholesterol-rich diet, i.e., possesses a hypolipidemic activity. These findings are consistent with the results of animal experiments in which hyperlipidemia was also produced by cholesterol-rich diet (Bordia et al., 1975; Jain, 1975; Kritchevsky, 1975; Jain and Konar, 1978; Bordia and Verma, 1980; Mand et al., 1985; Mirhardi et al., 1991; Heinle and Betz, 1994). This corresponds to the majority of clinical data indicate that garlic and garlic-derived preparations decrease serum levels of cholesterol and other lipids (Silagy and Neil, 1994; Warshafsky et al., 1993). The mean hypocholesterolemic effect in these studies was moderate (about 10%) but statistically significant, as well as in our experiments. It should be mentioned that some clinical studies did not revealed hypolipidemic effect of garlic (Isaacsohn et al., 1998). Recently, a number of studies were carried as well in order to evaluate the effects of garlic/garlic extracts on the blood lipid profile (Zeng et al., 2012; Ried et al., 2013; Sahebkar et al., 2015) and atherosclerotic lesion formation (Rafieian-Kopaei, 2013; Rafieian-Kopaei et al., 2014). The antioxidant properties of garlic, shown in WEHI-164 fibrosarcoma tumor growth in BALB/c mice (Shirzad et al., 2011) might be relevant to the effects of garlic in atherosclerosis (Rafieian-Kopaei, 2013; Rafieian-Kopaei et al., 2014).

In the *ex vivo* experiments we have shown that blood serum of rabbits treated for a long time with garlic preparation induced significantly lower cholesterol accumulation in cultured cells. This finding is in agreement with the results of clinical trials which have shown that the patients' serum obtained after garlic tablet administration induced significantly lesser cholesterol accumulation in cultured cells than serum taken before garlic administration (Orekhov et al., 1985; Orekhov et al., 1995).

We have also shown that garlic preparation inhibits the development of atherosclerosis that was induced in injured aorta by a cholesterol-rich diet, i.e., it exerts an anti-atherogenic effect *in vivo*. The anti-atherogenic properties of garlic preparation are consistent with the majority of data on the effect of garlic on experimental atherosclerosis (Bordia et al., 1975; Jain, 1975; Kritchevsky, 1975; Jain and Konar, 1978; Bordia and Verma, 1980). However, in the studies of Lee and Lee (1980) and Heinle and Betz (1994) garlic had no effect in a model similar to that used in our study (mechanical injury followed by cholesterol-rich diet). Such discrepancy may be explained by the effects of balloon catheterization used in those studies, which leads to the formation of partially endothelialized intimal thickening. Possibly, in this case the results may be also explained by the effects of the endothelium on lipid accumulation and the intimal thickening growth, but not only by the effect of garlic itself. In addition, those studies have been performed on pig model using raw garlic in combination with onion (Lee and Lee, 1980) and rat model using pure garlic powder (Heinle and Betz, 1994). It should be also stressed that we have tested garlic preparation with a prolonged action. Thus, the results of our *in vivo* study correspond well to the experimental data obtained in *in vitro* and *ex vivo* studies in the cellular model.

Concluding remarks

In addition to widely used mouse atherosclerosis models in the contemporary research, the use of rabbits for experimental models of atherosclerosis continues to contribute in our understanding the

mechanisms of atherogenesis (Xiangdong et al., 2011; Getz and Reardon, 2012; Kapourchali et al., 2014; Fan et al., 2015; Priyadharsini, 2015). In the present study we, utilizing a rabbit atherosclerosis model, evaluated effects of garlic preparation on serum atherogenicity and formation of intimal thickening and showed that garlic preparation can prevent the progression of already existing atherosclerotic lesions, and the emergence of the new ones, thus reducing the risk of atherosclerosis and atherosclerosis-related cardiovascular diseases.

Conflict of interest

The authors state that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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